Summary in Graph

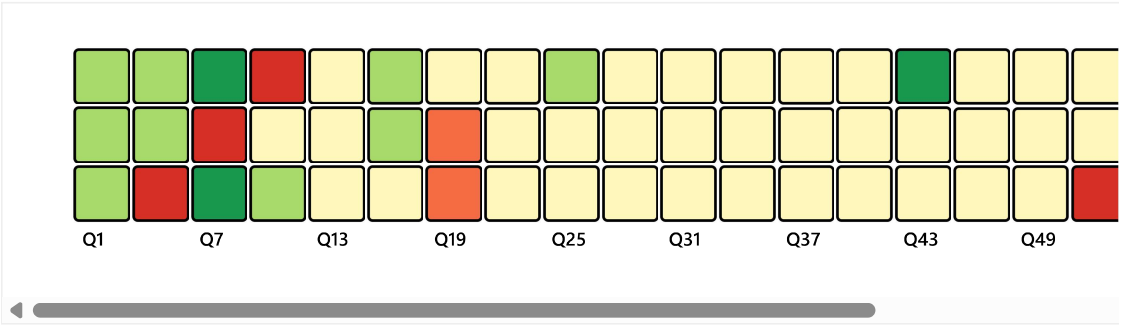
Exam Summary (GO Classes DA 2026 | All India Mock Test 2)

Qs. Attempted:	20 (11 + 9)	Correct Marks:	17 (9 + 8)
Correct Attempts:	13 (9 + 4)	Penalty Marks:	3.33 (0.67 + 2.67)
Incorrect Attempts:	7 (2 + 5)	Resultant Marks:	13.67 (8.33 + 5.33)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	180 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (32) out of 318 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

I don't know where she is. I suppose she _____ got stuck in traffic.

A. must have

B. should have

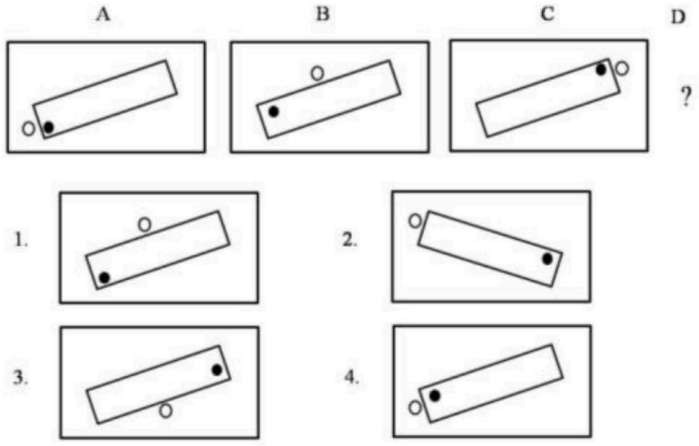
C. can have

D. might have

Your Answer: D | Correct Answer: D | Correct | Time taken: 00min 08sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Spatial Aptitude

What will be the next figure in the following sequence?

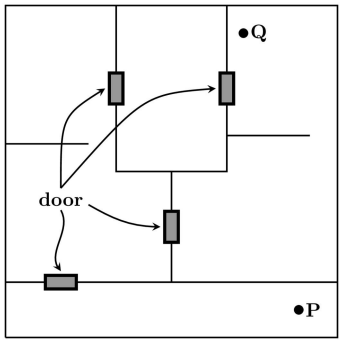


Your Answer: C | Correct Answer: C | Correct | Time taken: 00min 06sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

A building has several rooms and doors as shown in the top view of the building given below. The doors are closed initially.

What is the minimum number of doors that need to be opened in order to go from the point P to the point Q?



A. 4

B. 3

C. 2

D. 1

Your Answer: C | Correct Answer: C | Correct | Time taken: 00min 07sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

A sum of money is to be distributed among P, Q, R, and S in the proportion $5 : 2 : 4 : 3$, respectively.

If R gets ₹ 1000 more than S, what is the share of Q (in ₹)?

A. 500

B. 1000

C. 1500

D. 2000

Your Answer: D | Correct Answer: D | Correct | Time taken: 00min 05sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Both the numerator and the denominator of $\frac{3}{4}$ are increased by a positive integer, x, and those of $\frac{15}{17}$ are decreased by the same integer. This operation results in the same value for both the fractions.

What is the value of x?

A. 1

B. 2

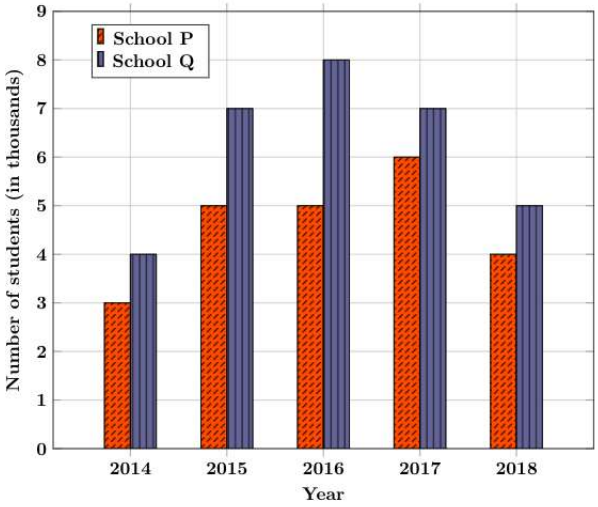
C. 3

D. 4

Your Answer: C | Correct Answer: C | Correct | Time taken: 00min 06sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

The following figure shows the data of students enrolled in 5 years $(2014 \text{ to } 2018)$ for two schools P and Q. During this period, the ratio of the average number of the students enrolled in school P to the average of the difference of the number of students enrolled in schools P and Q is _____.



A. $8 : 23$
B. $23 : 8$
C. $23 : 31$
D. $31 : 23$

Your Answer: A | Correct Answer: B | Incorrect | Time taken: 00min 10sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

To construct a wall, sand and cement are mixed in the ratio of $3 : 1$. The cost of sand and that of cement are in the ratio of $1 : 2$.

If the total cost of sand and cement to construct the wall is 1000 rupees, then what is the cost (in rupees) of cement used?

A. 400

B. 600

C. 800

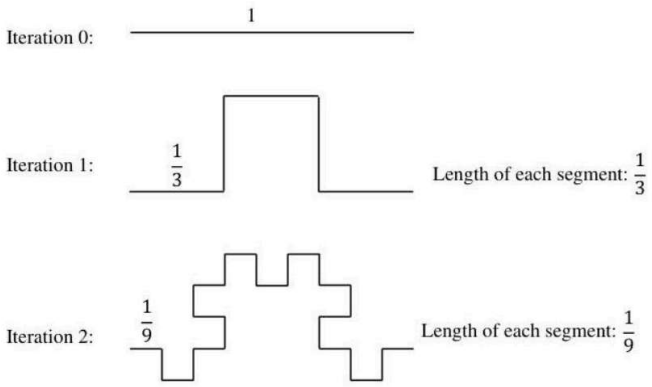
D. 200

Your Answer: A | Correct Answer: A | Correct | Time taken: 00min 48sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

The following figures show three curves generated using an iterative algorithm. The total length of the curve generated after 'Iteration n ' is:

Note: The figures shown are representative.



A. $\left(\frac{5}{3}\right)^{\frac{n}{2}}$
B. $\left(\frac{5}{3}\right)^{n}$
C. $\left(\frac{5}{3}\right)^{2n}$
D. $\left(\frac{5}{3}\right)^{n(2n-1)}$

Your Answer: D | Correct Answer: B | Incorrect | Time taken: 01min 28sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

Consider five people- Mita, Ganga, Rekha, Lakshmi, and Sana. Ganga is taller than both Rekha and Lakshmi. Lakshmi is taller than Sana. Mita is taller than Ganga.

Which of the following conclusions are true?

1. Lakshmi is taller than Rekha
2. Rekha is shorter than Mita
3. Rekha is taller than Sana
4. Sana is shorter than Ganga

A. 1 and 3
B. 3 only
C. 2 and 4
D. 1 only

Your Answer: C | Correct Answer: C | Correct | Time taken: 00min 04sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Spatial Aptitude

In the 4×4 array shown below, each cell of the first three columns has either a cross (X) or a number, as per the given rule.

1	1	2	
2	X	3	
2	X	4	
1	2	X	

Rule: The number in a cell represents the count of crosses around its immediate neighboring cells (left, right,

top, bottom, diagonals).

As per this rule, the **maximum** number of crosses possible in the empty column is

A. 0
B. 1
C. 2
D. 3

Your Answer: B | Correct Answer: C | Incorrect | Time taken: 00min 07sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be a linear transformation. Suppose

$\{v_1, w_1\}$ is a **linearly dependent** set of vectors and

$\{v_2, w_2\}$ is a **linearly independent** set of vectors, where

$$A(v_1) = v_2 \quad \text{and} \quad A(w_1) = w_2.$$

Which of the following statements is **correct**?

A. Such a linear transformation A always exists

B. Such a linear transformation A exists if $v_1 \neq w_1$

C. Such a linear transformation A exists only if $v_2 = cw_2$ for some scalar c

D. Such a linear transformation A can never exist

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be an $n \times m$ matrix whose entries belong to $\{+1, -1, 0\}$. Suppose that each column of A contains *exactly one* entry equal to $+1$ and *exactly one* entry equal to -1.

Which of the following conclusions can be drawn?

A. $\text{Rank}(A) \leq n - 1$

B. $\text{Rank}(A) = m$

C. $n \leq m$

D. $n - 1 \leq m$

Your Answer: A | Correct Answer: A | Correct | Time taken: 16min 42sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be a non-singular $n \times n$ matrix with real entries. Let $B = A^T$ denote the transpose of A.

Which of the following matrices is necessarily *positive definite*?

A. $A + B$
B. AB
C. $A^{-1} + B^{-1}$
D. ABA

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Minimum of dimension of the intersection of two seven dimensional vector subspaces in a twelve dimensional vector space is:

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Assume that $X \sim \text{Binomial}(n, p)$ for some $n \geq 1$ and $0 < p < 1$,
and $Y \sim \text{Poisson}(\lambda)$ for some $\lambda > 0$.

Suppose that

$$\mathbb{E}[X] = \mathbb{E}[Y]$$

Then:

A. $\text{Var}(X) = \text{Var}(Y)$
B. $\text{Var}(X) < \text{Var}(Y)$
C. $\text{Var}(X) > \text{Var}(Y)$
D. $\text{Var}(X)$ may be larger or smaller than $\text{Var}(Y)$, depending on the values of n, p, λ

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X and Y be independent and identically distributed random variables.

Consider the following statements:

S1: If X has a normal distribution, then the random variable $X + Y$ also has a normal distribution.

S2: If X has a binomial distribution, then the random variable $X + Y$ also has a binomial distribution.

Which of the following is **correct**?

A. Only statement S1 is correct.

B. Only statement S2 is correct.

C. Both statements S1 and S2 are correct.

D. Neither statement S1 nor S2 is correct.

Your Answer: C | Correct Answer: C | Correct | Time taken: 03min 36sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the following set of labeled examples in a two-dimensional input space:

- Class -1 : $(0,0),\ (-2,-2)$
- Class $+1$: $(1,3),\ (2,2),\ (3,-1)$

We train a hard-margin support vector machine (SVM) on this dataset.

Determine the separating line learned by the hard-margin SVM and find the margin in the input space.

What is the value of the hard margin?

A. $\sqrt{2}$

B. 2

C. $\dfrac{2}{\sqrt{5}}$

D. $\sqrt{5}$

Your Answer: D | Correct Answer: D | Correct | Time taken: 29min 11sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider a dataset with n training examples, where each example has a feature vector $x_i \in \mathbb{R}^d$ and a target value y_i. Let $X \in \mathbb{R}^{n \times d}$ be the data matrix whose i-th row is $x_i^\top$, and let $y \in \mathbb{R}^n$ be the target vector.

The following regularized least squares objective is given:

$$L(w) = \sum_{i=1}^{n} (y_i - x_i^\top w)^2 + \lambda \|w\|_2^2,$$

where $\lambda > 0$. Assume all required matrices are invertible. Which of the following is the closed-form solution for the optimal w?

A. $(X^\top X)^{-1} X y + \lambda I$

B. $2(X^\top X + \lambda I)^{-1} X^\top y$

C. $\lambda (X^\top X)^{-1} X^\top y$

D. $(X^\top X + \lambda I)^{-1} X^\top y$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Which of the following statements is TRUE regarding the application of logistic regression and linear regression on binary classification tasks?

A. Logistic regression models the probability of an instance belonging to a particular class, while linear regression predicts class labels directly.

B. Logistic regression is able to generate non-linear decision boundaries with respect to original input features when the data are not linearly separable.

C. Both logistic regression and linear regression have a closed-form solution for the optimal parameters.

D. For linearly separable data, logistic regression via gradient descent converges in a finite number of steps.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Which of the following statements is true about the K-means algorithm with the following objective function?

$$J = \sum_{n=1}^{N}\sum_{k=1}^{K} z_{nk}\|x_n - \mu_k\|^2 \text{ subject to } z_{nk} \in \{0,1\} \text{ and } \sum_{k=1}^{K} z_{nk} = 1 \text{ for all } n$$

A. The objective value might increase during training ($J_{t+1} > J_t$) due to non-convexity of the objective function or poor initialization.

B. K-means converges to a local minimum because the objective function is convex.

C. Increasing the number of clusters (K) always decreases (or keeps constant) the optimal objective value.

D. K-means is guaranteed to converge to the global minimum of the objective function.

Your Answer: B | Correct Answer: C | Incorrect | Time taken: 19min 22sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the following training examples for a perceptron.

Training Example	x_1	x_2	Class
a	0	1	-1
b	2	0	-1

Suppose the initial weights are

$$w_0 = -0.5, \quad w_1 = 0, \quad w_2 = 1,$$

and the learning rate is $\eta = 0.5$.

Using the perceptron learning algorithm, the model is trained on instance **(a)** only. What are the updated weights after this training step?

A. $w_0 = -1,\ w_1 = 0,\ w_2 = 0.5$

B. $w_0 = -1.5,\ w_1 = -0.5,\ w_2 = 0.5$

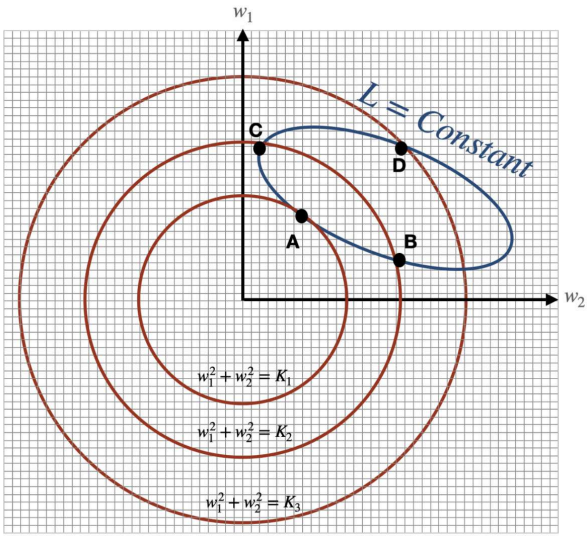
C. $w_0 = -1.5,\ w_1 = 0,\ w_2 = 0$

D. $w_0 = -2,\ w_1 = 0,\ w_2 = -0.5$

Your Answer: B | Correct Answer: A | Incorrect | Time taken: 02min 56sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the error function $E(\mathbf{w}) = L(\mathbf{w}) + \mathcal{R}(\mathbf{w})$, where $\mathcal{R}(\mathbf{w}) = w_1^2 + w_2^2$ denotes L2 regularization and $\mathbf{w} = [w_1, w_2]^\top \in \mathbb{R}^2$.



In the figure shown, the blue ellipse represents a level set of the data loss $L(\mathbf{w})$, i.e., $L(\mathbf{w})$ is constant for all points lying on the ellipse.

The red concentric circles correspond to level sets of the regularization term $w_1^2 + w_2^2$.
Points A, B, C, and D lie on the same blue ellipse.

Which of the following points has the **minimum value of the total error** $E(\mathbf{w})$?

A. A

B. B

C. C

D. D

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Let n be a positive integer.

Consider the two statements below:

- S1: If $f(n) > g(n)$ for all n then $g(n)$ is ALWAYS $o(f(n))$. where o is small-oh.
- S2: If $f(n) < g(n)$ for all n then $f(n)$ is ALWAYS $o(g(n))$. where o is small-oh.

Which of the following is the correct option?

A. $S1$ is correct but $S2$ is incorrect

B. $S1$ is incorrect but $S2$ is correct

C. Both are correct

D. Both are incorrect

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Let $f(n)$ and $g(n)$ be two positive functions defined for all positive integers n.

Consider the following statements:

- **S1:** If $f(n) > g(n)$ for all n, then $g(n) = o(f(n))$, where $o(\cdot)$ denotes little-oh.
- **S2:** If $f(n) = o(g(n))$, then there exists a positive integer n_0 such that $f(n) < g(n)$ for every $n \geq n_0$.

Which of the following is correct?

A. S1 is correct but S2 is incorrect
B. S1 is incorrect but S2 is correct
C. Both S1 and S2 are correct
D. Both S1 and S2 are incorrect

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Let $G = (V, E)$ be a **connected**, undirected graph with **non-negative edge weights**. Let $s \in V$ be a source vertex, and suppose the shortest paths from s to all other vertices in V have been computed.

Now, the weight of **every edge** in the graph is increased by 1.

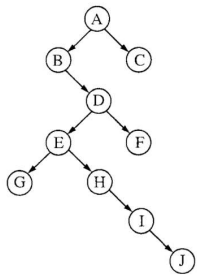
Which of the following statements is **always true**, regardless of the structure of G?

A. All shortest paths from s to every other vertex remain unchanged.
B. Some shortest paths from s to other vertices may change.
C. The shortest path distances from s to all vertices increase by the same amount.
D. The shortest path tree rooted at s remains unchanged.

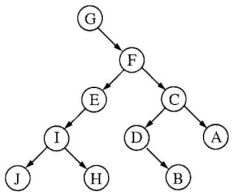
Your Answer: B | Correct Answer: B | Correct | Time taken: 04min 55sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

If $Tree1$ and $Tree2$ are the trees indicated below,



Tree1



Tree2

which traversals of $Tree1$ and $Tree2$, respectively, will produce the same sequence of node names?

A. Preorder, postorder

B. Postorder, inorder

C. Postorder, postorder

D. Inorder, inorder

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures

A full binary tree is a rooted tree in which every internal node has exactly two children. How many internal nodes are there in a full binary tree with 500 leaves?

Your Answer: | Correct Answer: 499 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The function $f : [-1, 1] \to \mathbb{R}$ defined by $f(x) = 1 - x^2$ has

A. no local maxima or minima in $(-1, 1)$.

B. has exactly one local maximum and two local minima in $(-1, 1)$

C. has exactly one local maximum in $(-1, 1)$.

D. has exactly one local minimum in $(-1, 1)$.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Consider the following constrained optimization problem:

$$\text{Maximize } f(x, y) = xy$$
$$\text{subject to } 15x + 20y = 1000.$$

Assume that the Lagrange multiplier satisfies $\lambda < 0$.

Which of the following is the **correct Lagrangian function** for this problem?

A. $L(x, y, \lambda) = x + y - \lambda(15x + 20y - 1000)$

B. $L(x, y, \lambda) = x + y + \lambda(15x + 20y - 1000)$

C. $L(x, y, \lambda) = xy + \lambda(15x + 20y - 1000)$

D. $L(x, y, \lambda) = xy - \lambda(15x + 20y + 1000)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The value of a function at $x = 5$ is $f(5) = 3.2$, and the value of its derivative at $x = 5$ is $f'(5) = 1.2$. Using **first-order approximation**, what is the approximate value of the function at $x = 5.2$?

A. 2.96

B. 3.08

C. 3.44

D. 3.92

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in Python

Consider the following Python program:

```
a = [10, 20, 30, 40, 50]
b = a[1:4]
b[0] = 99
print(a)
```

What will be printed?

A. `[10,99,30,40,50]`

B. `[10,20,30,40,50]`

C. `[99,30,40]`

D. It raises a runtime error

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Numerical Type | Award: 1 | Penalty: 0 | Programming in Python

Consider the following Python program:

```
def f(n):
    if n == 0:
        return 0
    if n % 2 == 0:
        return f(n - 1)
    return 1 + f(n - 1)

print(f(7))
```

What will be printed?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

We have a relation $\mathrm{R(A, B, C, D)}$. In each of the following questions, you are given a set F of functional dependencies, and a decomposition of R into two tables $R1$ and $R2$. In each case, determine whether the decomposition is lossy or lossless.

- $\mathrm{S1 : F = \{BC \rightarrow D, C \rightarrow D\}}$; decompose into $R1(A, B, C)$ and $R2(B, D)$.
- $\mathrm{S2 : F = \{}B \rightarrow C, C \rightarrow A, AC \rightarrow D\}$; decompose into $R1(A, B, C)$ and $R2(B, D)$.

Which of the following is correct?

A. $S1$ is a lossless decomposition and $S2$ is a lossless decomposition.

B. $S1$ is a lossless decomposition and $S2$ is a lossy decomposition.

C. $S1$ is a lossy decomposition and $S2$ is a lossless decomposition.

D. $S1$ is a lossy decomposition and $S2$ is a lossy decomposition.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Assume that a data file has an index consisting of N items, where N is large. If a binary search of the index is used to find an item, then, of the following, which best approximates the mean number of comparisons required to locate a specific index entry?

A. $(\mathrm{N} + 1)/2$

B. $\mathrm{N}(\mathrm{N} - 1)/2$

C. $(\log_2 \mathrm{N}) - 1$

D. $\mathrm{N} \log_2 \mathrm{N}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Consider some convex function $f : \mathbb{R}^d \to \mathbb{R}$, and assume that f is twice-differentiable. What can we say about $\nabla_x^2 f(x) \in \mathbb{R}^{d \times d}$, the Hessian of $f(x)$? (Hint: think about the case when $d = 1$, and what the second derivative of f says about its shape.)

A. $\nabla_x^2 f(x)$ is negative semi-definite.

B. $\nabla_x^2 f(x)$ is negative definite.

C. $\nabla_x^2 f(x)$ is positive definite.

D. $\nabla_x^2 f(x)$ is positive semi-definite.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let A and B be 2×2 matrices whose entries belong to $\{0, 1\}$.

All arithmetic is performed *modulo 2*, that is, $1 + 1 = 0$.

Suppose that $\mathrm{tr}(A) = \mathrm{tr}(B) \quad \text{and} \quad \mathrm{tr}(A^2) = \mathrm{tr}(B^2)$.

Which of the following statements is/are necessarily true?

A. A and B have the same eigenvalues
B. A and B have the same determinant
C. A and B may have different eigenvalues
D. $A = B$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let

$$A = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \qquad Q(x) = x^T A x \quad \text{for } x \in \mathbb{R}^3.$$

Which of the following statements is/are **TRUE**?

A. A has exactly two positive eigenvalues.

B. All the eigenvalues of A are positive.

C. $Q(x) \geq 0$ for all $x \in \mathbb{R}^3$.

D. $Q(x) < 0$ for some $x \in \mathbb{R}^3$.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

For any $n \times n$ matrix B, let $N(B) = \{ x \in \mathbb{R}^n \mid Bx = 0 \}$ denote the null space of B.

Let A be a 4×4 matrix such that

$$\dim\Big(N(A-2I)\Big) = 2,$$
$$\dim\Big(N(A-4I)\Big) = 1.$$

and $\text{rank}(A) = 3$.

Then, which of the following statements is/are true?

A. $0, 2$, and 4 are eigenvalues of A.

B. $\det(A) = 0$.

C. A is not diagonalizable.

D. $\text{trace}(A) = 8$.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

Let

$$S = \left\{ (x_1, x_2, x_3, \ldots, x_{50}) \in \mathbb{R}^{50} \;\middle|\; \begin{array}{l} x_1 = x_2 = \cdots = x_{25} = 0, \\ x_{26} + x_{27} + \cdots + x_{50} = 0 \end{array} \right\}.$$

The dimension of the vector space S is:

Your Answer: | Correct Answer: 24 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let S and T be vector subspaces of a vector space V. Then $S \cup T$ is a subspace of V

A. is never true

B. if and only if one of S or T is trivial

C. if and only if $S \subseteq T$ or $T \subseteq S$

D. if and only if $S \cap T$ is a nonzero vector space.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Suppose A is a 3×3 symmetric matrix such that

$$[x, y, 1]A\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = xy - 1$$

Let p be the number of positive eigenvalues of A and let $q = \text{rank}(A) - p$. Then

A. $p = 1$.

B. $p = 2$.

C. $q = 2$.

D. $q = 1$.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X and Y be two independent random variables, each representing the number of failures preceding the first success in a sequence of Bernoulli trials, with p as the probability of success in a single trial and q as the probability of failure. Then $P(X = Y)$ is:

A. $\frac{q}{1+q}$

B. $\frac{p}{1+p}$

C. $\frac{q}{1+p}$

D. $\frac{p}{1+q}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Given the probability density function

$$f(x) = \begin{cases} e^{-x}, & x \geq 0, \\ 0, & x < 0, \end{cases}$$

find the value of the cumulative distribution function $F(2)$.

A. $1 + e^{-2}$

B. e^{-2}

C. e^{-1}

D. $1 - e^{-2}$

Your Answer: D | Correct Answer: D | Correct | Time taken: 04min 36sec | Discuss

Q #34 | Numerical Type | Award: 2 | Penalty: 0 | Probability

Let X and Y have joint probability density function given by

$$f_{X,Y}(x,y) = \begin{cases} 2, & 0 \le x \le 1-y, \quad 0 \le y \le 1 \\ 0, & \text{otherwise .} \end{cases}$$

If f_Y denotes the marginal probability density function of Y, then $f_Y(1/2) =$ _____

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X and Y be two random variables having the joint probability density function

$$f(x,y) = \begin{cases} 2 & \text{if } 0 < x < y < 1 \\ 0 & \text{otherwise .} \end{cases}$$

Then the conditional probability $P\left(X \le \frac{2}{3} \middle| Y = \frac{3}{4}\right)$ is equal to

A. $\frac{5}{9}$

B. $\frac{2}{3}$

C. $\frac{7}{9}$

D. $\frac{8}{9}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Let $X_1, X_2, \ldots, X_n$ be independent exponential random variables with parameters $\lambda_1, \lambda_2, \ldots, \lambda_n$.

Define $Y = \min(X_1, X_2, \ldots, X_n)$.

The random variable Y has an exponential distribution with parameter λ.

Which of the following statements is/are correct?

A. $P(Y > t) = P(X_1 > t, X_2 > t, \ldots, X_n > t)$

B. $P(Y > t) = P(X_1 > t)\, P(X_2 > t) \cdots P(X_n > t)$

C. $\lambda = \lambda_1 + \lambda_2 + \cdots + \lambda_n$

D. $\lambda = \max\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$

Your Answer: B;D | Correct Answer: A;B;C | Incorrect | Time taken: 03min 57sec | Discuss

Q #37 | Numerical Type | Award: 2 | Penalty: 0 | Probability

Let $\mathbf{X} \sim \mathcal{N}_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, i.e., $\mathbf{X} = (X_1, X_2, X_3)^\top$ is a three-dimensional multivariate normal random vector with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$.

The mean vector is

$$\boldsymbol{\mu} = (1,\ 1,\ 1),$$

and the covariance matrix is

$$\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & c \\ 1 & c & 2 \end{pmatrix}.$$

Find the value of c such that the random variables X_2 and $-X_1 + X_2 - X_3$ are independent

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Consider a model $\mathcal{H}_\alpha$ whose complexity is controlled by a hyperparameter α. We test two values of this hyperparameter: α_1 and α_2.

Let h_{α_1} and h_{α_2} be the hypotheses obtained after training using α_1 and α_2, respectively. Let D_{tr} denote the training dataset and D_{te} denote the test dataset. Let $E(h \mid D)$ denote the empirical error of hypothesis h on dataset D.

Suppose the following holds:

$$E(h_{\alpha_1} \mid D_{te}) - E(h_{\alpha_1} \mid D_{tr}) \ < \ E(h_{\alpha_2} \mid D_{te}) - E(h_{\alpha_2} \mid D_{tr})$$

What can we conclude from this?

A. Model $\mathcal{H}_{\alpha_1}$ is underfitted

B. Model $\mathcal{H}_{\alpha_2}$ is overfitted

C. The optimal model corresponds to a hyperparameter value between α_1 and α_2

D. Model $\mathcal{H}_{\alpha_1}$ is of a lower complexity than model $\mathcal{H}_{\alpha_2}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

We have 1500 labeled examples. On this set, we train and evaluate the SVM algorithm, optimizing the hyperparameters C and γ. For evaluation, we use nested cross-validation with 5 folds in the outer loop and 5 folds in the inner loop. This means that for each combination of hyperparameter values C and γ we train five models. These models are trained on training sets that are not mutually disjoint: each pair of trained models share a certain number of training examples. Calculate how many training examples are shared by each pair of models trained in the inner loop of nested cross-validation. **How much would this number increase if we increase the number of folds in the inner loop to 10 ?**

A. 288

B. 120

C. 150

D. 240

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

We want to learn a **Naive Bayes classifier** to predict whether a person is **Happy** using two inputs: **Age** (a real number) and **Occupation** (a category).

For the variable *Age*, assume the following model. For each value of *Happy*, the probability density of Age is uniform between two numbers L and H:

$$p(\text{Age} = x \mid \text{Happy}) = \begin{cases} \dfrac{1}{H-L}, & L \le x \le H, \\ 0, & \text{otherwise.} \end{cases}$$

The values of L and H are estimated from the training data (minimum and maximum Age values for each class).

The training data is:

Age	Occupation	Happy
20	CTO	No
40	Professor	No
50	CTO	Yes
30	Professor	Yes
50	Professor	Yes

After learning the classifier, find

$$P(\text{Happy} = \text{Yes} \mid \text{Occupation} = \text{Professor},\ \text{Age} = 36).$$

A. $\frac{1}{3}$

B. $\frac{1}{2}$

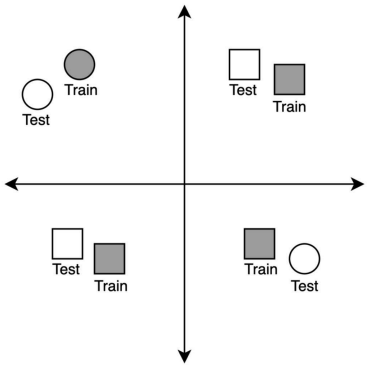
C. $\frac{2}{3}$

D. $\frac{3}{4}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Consider the dataset pictured below. The features of each datapoint are given by its position. So the datapoint $(0, 1)$ appears at position $(0, 1)$. The ground truth label of the datapoint is given by its shape, either circle or square. You have a test set of datapoints, shown with no fill, and a train set of data, shown with a grey fill. Which of the following statements are true about classifying this data?



A. It is possible for a linear SVM to have 100% train accuracy

B. It is possible for a linear SVM to have 100% test accuracy

C. KNN with $K = 1$ has higher test accuracy than with $K = 4$

D. KNN with $K = 1$ has higher train accuracy than with $K = 4$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

In a network graph $G = (V, E)$, each edge $e \in E$ has a probability p_e of successful transmission. Given two vertices $s \neq t \in V$, your goal is to find the s - t path with the **highest probability of successful transmission.**

To achieve this, you construct a new graph $G' = (V', E')$ with $V' = V$ and $E' = E$, and assign weights to edges in G' so that finding the shortest path in G' **using Dijkstra's algorithm** will correspond to finding the **highest probability path in** G.

Which of the following could be appropriate weights for edge e in G' if its probability in G is p_e? (Select all that apply.)

A. p_e

B. $\log(p_e)$

C. $\log\left(\frac{1}{p_e}\right)$ or $-\log(p_e)$

D. p_e^2

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Professor Sortun uses the following algorithm to merge k **sorted lists**, each having n/k elements.

She merges the lists sequentially as follows: first the **first and second lists**, then the resulting list with the **third list**, then with the **fourth list**, and so on, until all lists are merged into a single sorted list of n elements.

Assume that **merging two sorted lists of sizes** a **and** b **takes** $\Theta(a + b)$ **time.**

What is the **worst-case running time** of this algorithm in terms of n and k ?

A. $\Theta(n \log k)$

B. $\Theta(n \log n)$

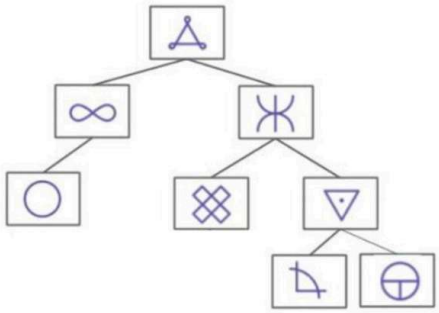
C. $\Theta(kn)$

D. $\Theta(n + k)$

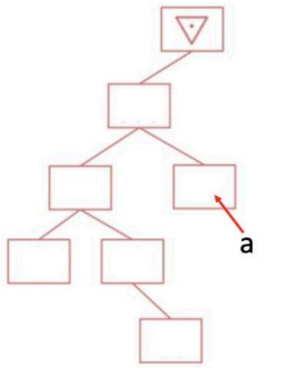
Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider the following Binary search tree, some objects and the empty structured binary search tree shown. Assume no two objects are equal.



Binary Search Tree (Filled)



Binary Search Tree (Empty Structure)

What could be the possible object(s) in place of **"a"** as indicated in the above figure?

A. 

B. 

C. 

D. 

Your Answer: C | Correct Answer: B | Incorrect | Time taken: 26min 29sec | Discuss

Q #45 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider the following singly linked list where the head points to the first node of the linked list.

$$2 \rightarrow 4 \rightarrow 6 \rightarrow 8 \rightarrow 10 \rightarrow \text{NULL}$$

If we execute the following code on the above-linked list, then what will be the final linked list?

```
    def recursive_swap(head):
        if head is None or head.next is None:
            return head

 5.     first = head
        second = head.next

        first.next = recursive_swap(second.next)
        second.next = first
10.
        return second
```

A. $2 \rightarrow 4 \rightarrow 6 \rightarrow 8 \rightarrow 10 \rightarrow$ NULL
B. $4 \rightarrow 2 \rightarrow 8 \rightarrow 6 \rightarrow 10 \rightarrow$ NULL
C. $2 \rightarrow 6 \rightarrow 4 \rightarrow 8 \rightarrow 10 \rightarrow$ NULL
D. $4 \rightarrow 2 \rightarrow 6 \rightarrow 8 \rightarrow 10 \rightarrow$ NULL

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider the following statements about an **undirected connected graph** with edge weights.

Which of the following statements are **TRUE**?

A. The shortest path between any two vertices always lies on some minimum spanning tree (MST).

B. If all edges of a graph have distinct weights, then the minimum spanning tree is unique.

C. If all edges of a graph have distinct weights, then the second-best minimum spanning tree is unique.

D. If all edges of a graph have distinct weights, then the maximum spanning tree is unique.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function such that

- $f(a) \geq b$
- $f(b) \leq a$

Then there exists a point $x \in [a, b]$ such that

A. $f(x) = x$
B. $f(x) = 0$
C. $f'(x) = 0$
D. $f''(x) = 0$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If objective function which is to be minimised is $f(x, y, z) = x + z$ and the constrained equation is $g(x, y, z) = x^2 + y^2 + z^2 = 1$. The point where minimum value occurs will be

A. $\left(\frac{1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}}\right)$
B. $\left(\frac{-1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$
C. $\left(\frac{-1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}}\right)$
D. $\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Consider a six-node Bayes network corresponding to the following factorization:

$$P(u, v, w, x, y, z) = P(u \mid w, x)P(v \mid w, z)P(w \mid x)P(x)P(y \mid x, z)P(z \mid u)$$

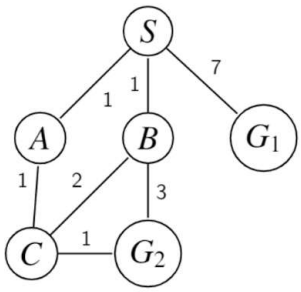
Use the d-separation method to test the conditional independence of variables x and z when conditioned on the remaining four variables. Examine which of the remaining four variables should be observed and which unobserved to make the variables x and z d-separated. **Which of the remaining four variables may be both observed and unobserved, if the variables x and z are d-separated?**

A. u
B. v
C. w
D. y

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Which goal is reached and what is the total cost of the solution found for the following state-space graph when using Breadth-First Search and Uniform-Cost Search (S is the start state, $\mathrm{G1}$ and $\mathrm{G2}$ are the goal states, arcs are bidirectional, no repeated state checking, break any ties alphabetically)?



A. BFS: G1 (Cost: 7), UCS: G2 (Cost: 4)

B. BFS: G2 (Cost: 4), UCS: G1 (Cost: 7)

C. BFS: G2 (Cost: 4), UCS: G2 (Cost: 4)

D. BFS: G1 (Cost: 7), UCS: G2 (Cost: 3)

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Consider the following statements about Gibbs sampling:

- **S1:** When using Gibbs sampling, to update a particular hidden variable, we compute the *most likely value* of that variable conditioned on all the others.
- **S2:** When using Gibbs sampling, successive samples (for example, $\theta^{(t)}$ and $\theta^{(t+1)}$) are independent.

Which of the following options is correct?

A. Both S1 and S2 are True

B. S1 is True and S2 is False

C. S1 is False and S2 is True

D. Both S1 and S2 are False

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

The island of Wantuutrewan is inhabited by exactly two types of natives: knights who always tell the truth and knaves who always lie.

Every native is a knight or a knave, but not both. You meet four islanders Aiken, Benny, Candy and Dueet. Two of them speak.

Aiken says: Benny never tells the truth.

Aiken says: Candy always lies.

Dueet says: Candy is a knave and I am a knave.

Which of the following is true?

A. Both Benny and Candy are knights.

B. Benny is a knight and Candy is a knave.

C. Benny is a knave and Candy is a knight.

D. Both Benny and Candy are knaves.

Your Answer: A | Correct Answer: A | Correct | Time taken: 04min 57sec | Discuss

Q #53 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Given a predicate P, and a domain D, we define the **truth set** of P to be the set of elements x in D for which $P(x)$ is true. The truth set of $P(x)$ is denoted by $\{x \in D \mid P(x)\}$.

Similarly, the **falsity set** of a predicate is the set of elements in D for which the predicate is false.

Let P and Q be two predicates, and let their truth sets be denoted as T_P, T_Q, and falsity sets be denoted as F_P, F_Q.

Which of the following identities are correct?

A. $T_P \cap T_Q = T_{P \wedge Q}$

B. $T_P \cup T_Q = T_{P \vee Q}$

C. $F_P \cap F_Q = F_{P \vee Q}$

D. $F_P \cup F_Q = F_{P \wedge Q}$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider a relation $\mathrm{R}(\mathrm{A}, \mathrm{B})$ that contains 100 tuples, and a relation $\mathrm{S}(\mathrm{B}, \mathrm{C})$ that contains 80 tuples; Make no assumptions about keys & assume that the domain of all attributes is integers. Consider the following relational algebra expression:

$$\pi_{\mathrm{B}}\mathrm{R} - (\pi_{\mathrm{B}}\mathrm{R} - \pi_{\mathrm{B}}\mathrm{S})$$

Let m be the minimum and M be the maximum number of tuples that could be in the result of the above expression, then $m + \mathrm{M}$?

Your Answer: | Correct Answer: 80 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider the following table whose name is T:

A	B	C
1	10	100
2	10	10
3	40	100
4	30	200
5	25	90

The number of rows in the output of the following SQL query is?

```
select A from T as R
where not exists (
    select * from T
    where T.B > R.B and T.C > R.C)
```

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Copyright & Stuff